

Global Renewable Energy in 2025: How Far Have We Come?

A Data-Driven Review of the Global Energy Transition

Tracking renewable progress across primary energy, electricity generation, and technology breakthroughs

Our World in Data Ember · Energy Institute

Fossil fuels still drive ~75% of global emissions — and cause 5 million+ premature deaths annually

75% of Global GHG Emissions Still Come from Fossil Fuels

 **75%**

of global GHG emissions

come from burning fossil fuels for energy

Source: Our World in Data – Renewable Energy

- Primary driver of climate change since the Industrial Revolution

 **5M+**

premature deaths per year

from fossil fuel air pollution globally

Source: Our World in Data – Renewable Energy

- Beyond CO₂: local air quality crisis demanding urgent action

 **~200 yrs**

of fossil fuel dominance

Since the Industrial Revolution, most nations' energy systems rely on fossil fuels

- Energy systems inertia makes rapid change challenging

 **Nuclear + Renewables**

= The pathway forward

Rapid shift to low-carbon sources is essential to cut CO₂ and end air pollution

- Renewables: solar, wind, hydro, geothermal, bioenergy

Note: Renewable energy includes solar, wind, hydropower, geothermal, bioenergy, wave and tidal. Traditional biomass is excluded from modern renewable statistics.

Only ~14% of global primary energy comes from renewables — electricity leads, transport and heating still lag

Renewables Now Supply ~14% of Global Primary Energy



Fig. 1: Share of primary energy from renewables by country, 2024 (substitution method) | Source: Our World in Data

 ~14%

Global primary energy from renewables

2024 data; substitution method
~1/7th of all energy consumed globally

Leaders

Norway · Brazil · New Zealand

Derive large shares of primary energy from renewables (hydro-rich nations)

- Abundant hydro resources key advantage

Why only 14%?

Primary energy = electricity + transport + heating

Transport and heating are harder to decarbonize; still reliant on oil and gas

- Electricity share (~30%) far exceeds total primary

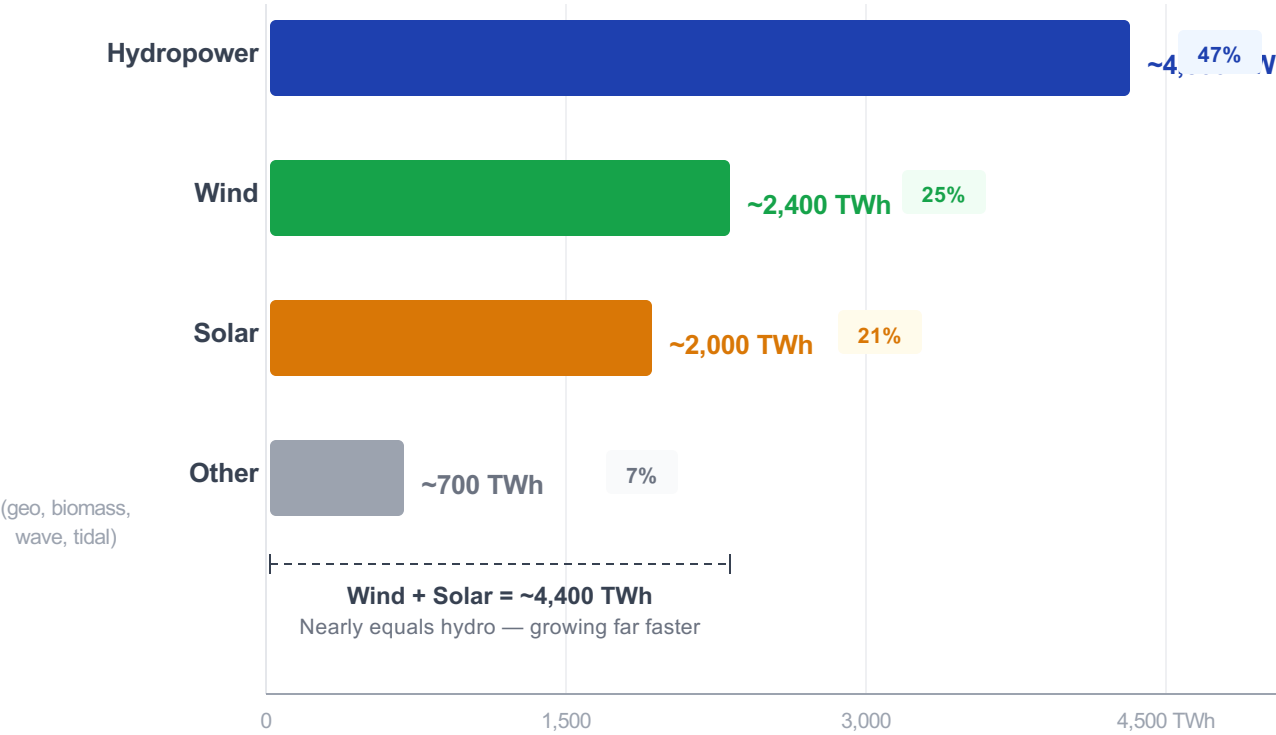
Laggards: Fossil-fuel-producing nations in the Middle East and parts of Asia remain below 5% renewable primary energy share.

Wind + solar combined (~4,400 TWh) now nearly rival hydro's ~4,500 TWh — and are growing far faster

Total renewable electricity: ~9,600 TWh globally (2024)

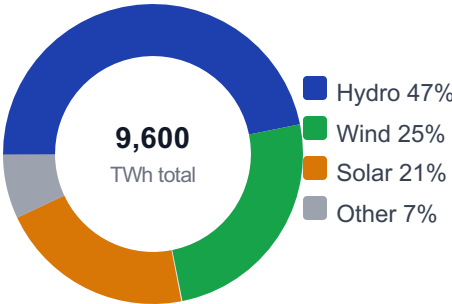
Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Global Renewable Electricity Generation by Technology (TWh, 2024)



Technology Breakdown — Renewable Electricity (2024)

Technology	Generation (TWh)	Share
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other Renewables	~700	~7%
Total Renewable	~9,600 TWh	100%



Strategic Implication

While hydropower still leads in total output, wind and solar are growing dramatically faster. Solar generation has surged since 2010 on falling costs and policy support.

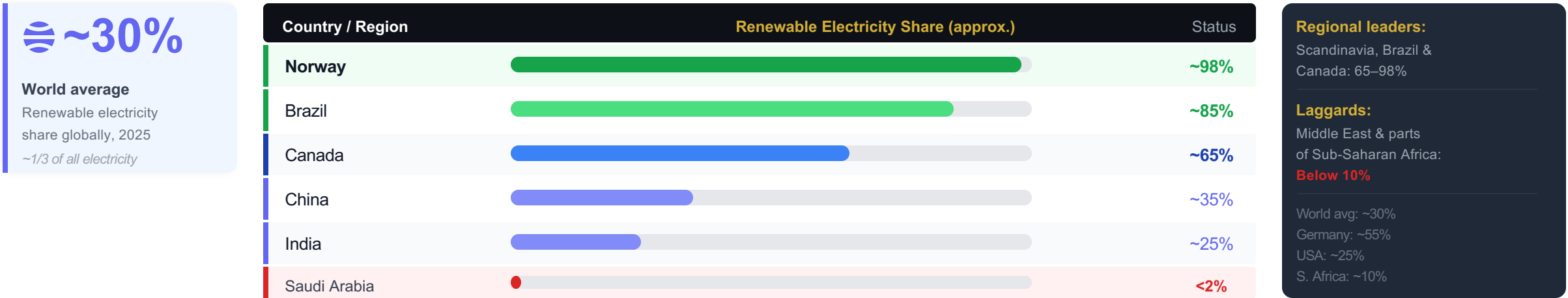
Electricity is renewables' best-performing sector: ~30% globally, up to 98% in Norway

Yet Saudi Arabia and much of Sub-Saharan Africa remain below 2–10% — a stark global divide

Almost One-Third of Global Electricity Now Comes from Renewables



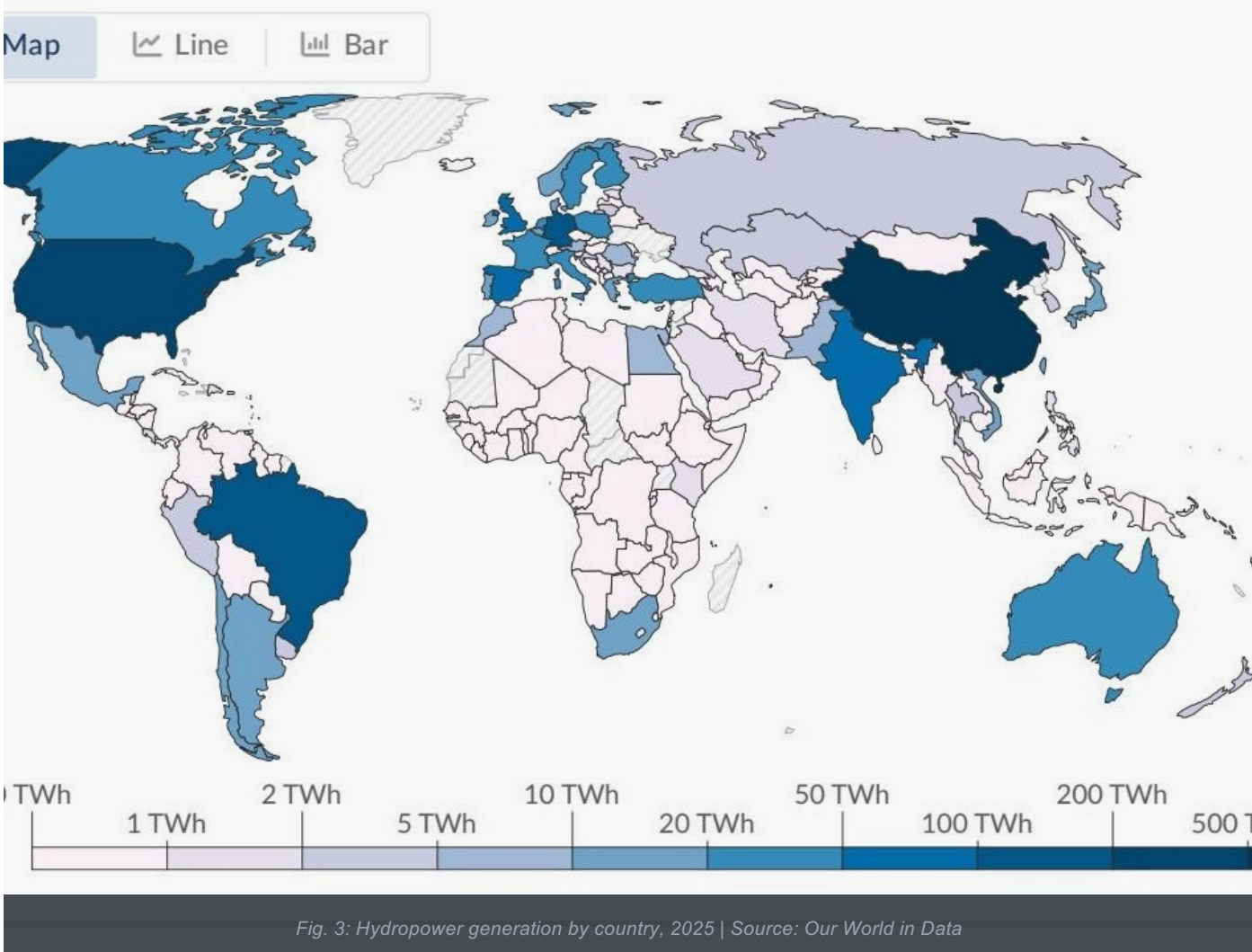
Fig. 2: Share of electricity from renewables by country, 2025 | Source: Our World in Data



Source: Our World in Data; Ember (2026). Country data approximate.

Hydropower still generates ~half of all renewable electricity — but its growth potential is limited vs. wind and solar

Hydropower Remains the Largest Renewable Source at ~50% of Generation



4,500
TWh generated

Global hydropower output (2024)
Approx. global figure

~47%
of renewable electricity
from hydropower

Top Producing Nations

#1

China

Largest producer
by total output

#2

Brazil

~85% electricity
from renewables

#3

Canada

~65% electricity
from renewables

Why Hydropower Matters

- Operating for **over 100 years** at scale — most mature renewable
- Provides reliable **baseload power**, unlike intermittent wind/solar
- Zero-carbon dispatchable electricity — critical grid resource

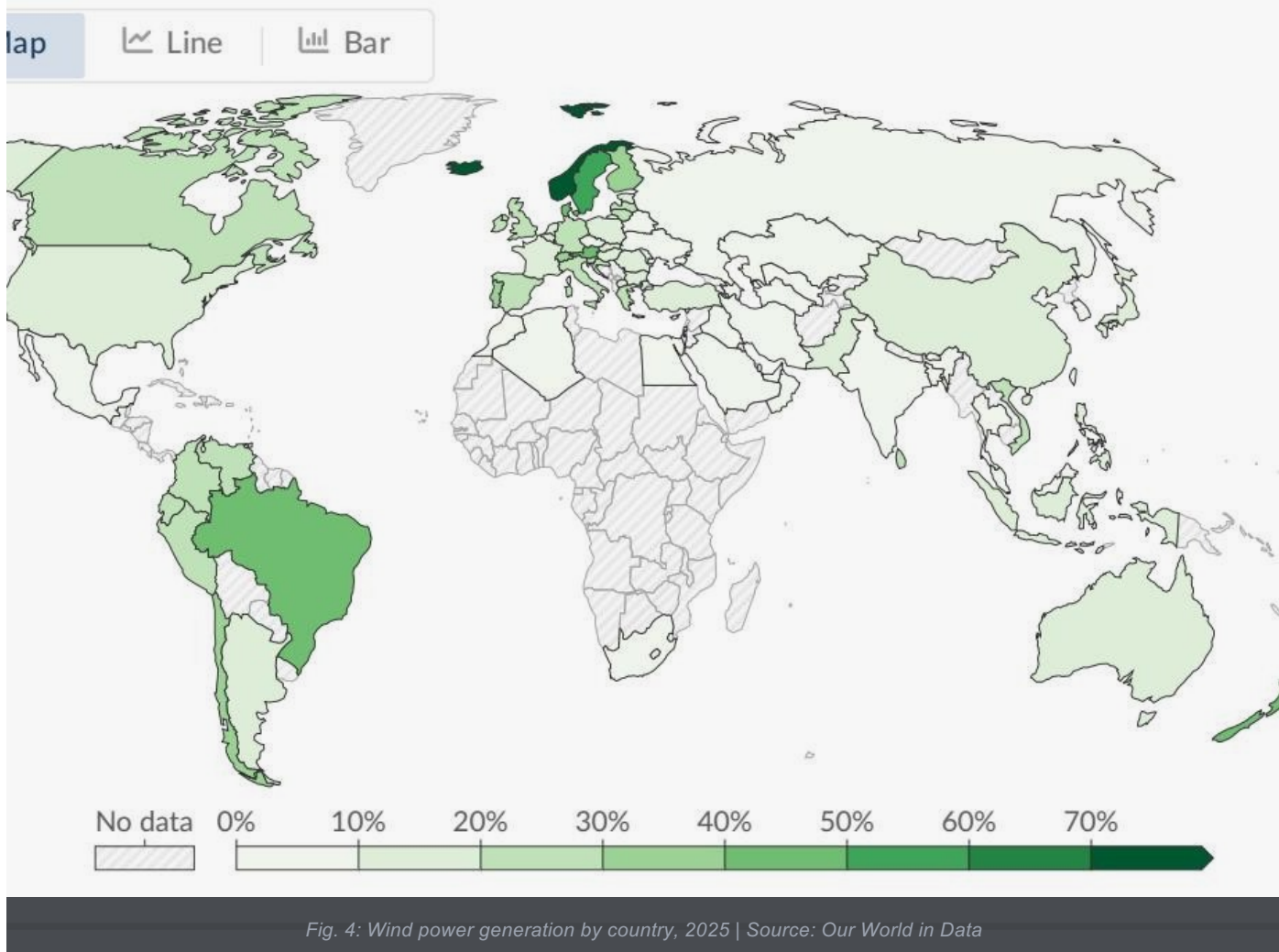
Growth constraint

Limited suitable new large-scale sites remain globally.
Future renewable growth must come from **wind and solar**.
Environmental concerns also constrain large dam construction

Solar is history's fastest-growing energy technology; wind + solar together now nearly rival hydropower

Driven by plummeting costs and policy support — growth dramatically accelerated since 2010

Wind and Solar Are the Fastest-Growing Energy Technologies in History



Wind Energy

~2,400

TWh generated globally (2024)

~25%

of renewable electricity

Includes onshore + offshore · Key regions: Europe, China, USA



Solar Energy

~2,000

TWh generated globally (2024)

~21%

of renewable electricity

Dramatic growth since 2010 · Driven by falling costs + policy

Combined Output:

Wind + Solar ≈ 4,400 TWh

vs. Hydro ~4,500 TWh — nearly equal, growing far faster

Growth Drivers



Solar costs fell 90%+ over the past decade, making it the cheapest electricity source in history

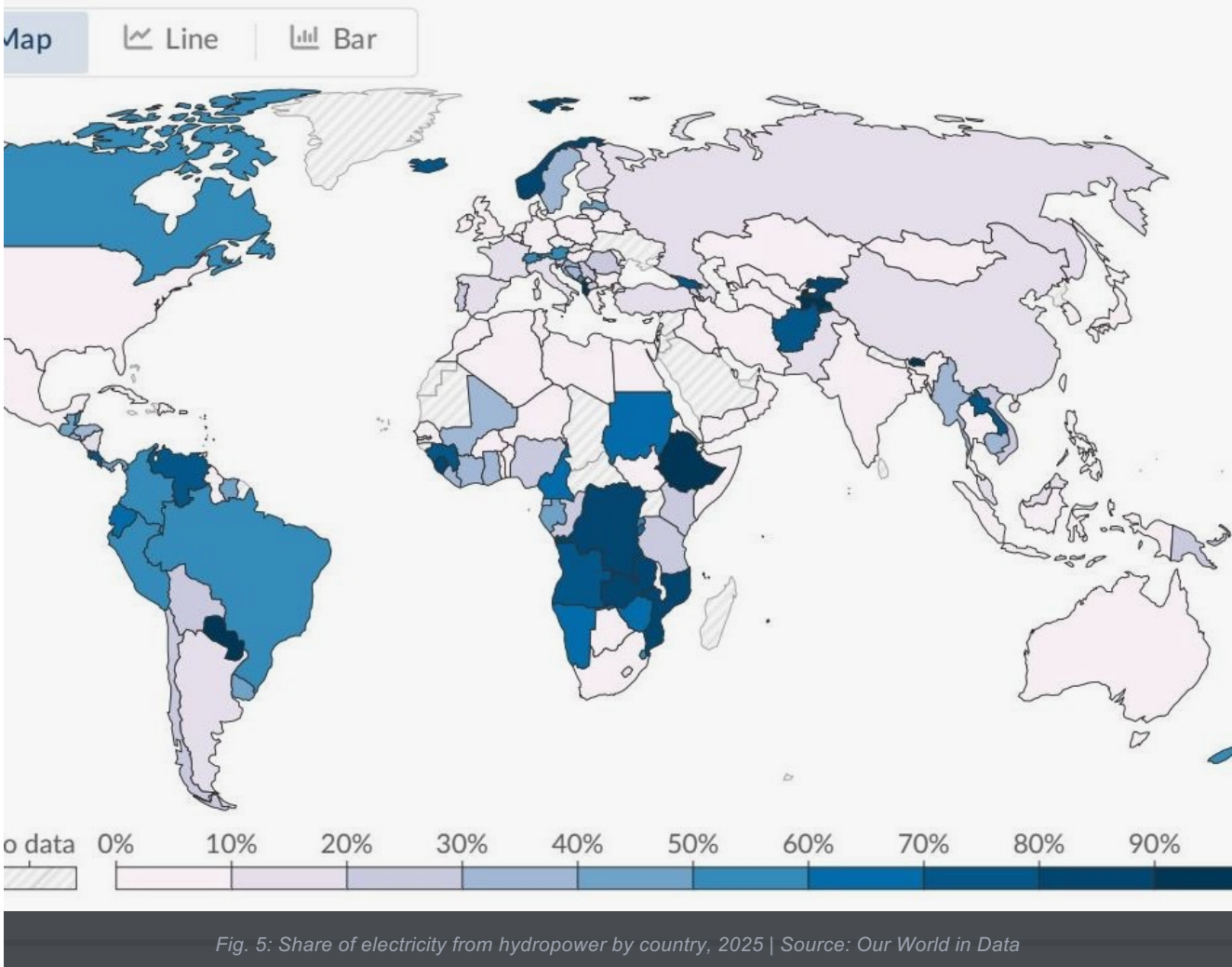


Policy support: Paris Agreement commitments, national renewable energy targets



Offshore wind expansion accelerating across Europe, Asia, and North America

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



Leaders: 60–98% Renewable Electricity

- Norway (~98%)

Abundant hydropower resources
- Brazil (~85%)

Amazon basin hydropower dominance
- Canada (~65%)

Hydro-rich geography + wind growth
- Germany (~55%)

Heavy investment in wind and solar
- Denmark surpasses 60%+ from wind alone

Mid-Tier: 25–40% Renewable Electricity

- China (~35%) · USA (~25%) · India (~25%)

Significant scale of investment; growing rapidly from large base

China is world's largest absolute renewable generator despite 35% share

Laggards: Below 10% Renewable Electricity

- Middle East

Saudi Arabia (<2%), UAE, Kuwait — fossil fuel exporters with minimal renewable investment; almost entirely oil and gas dependent
- Sub-Saharan Africa

South Africa (~10%) is exception; many nations have minimal renewable electricity infrastructure despite abundant solar potential

Access gap = energy justice and development imperative

Limited investment flows + grid infrastructure deficit compound the challenge

Renewables Are Growing Fast — But the Transition Must Accelerate

 **14% / 30%**

Renewable primary energy / electricity

A significant milestone — approximately one-seventh of all energy consumed globally now comes from renewables.

Still far short of 2050 climate targets
Most scenarios require 60%+ by 2030

01

 **~47% from Hydro**

Hydropower remains dominant

Hydropower generates roughly half of all renewable electricity. Top producers: China, Brazil, and Canada.

Growth potential limited; new hydro sites scarce
Future growth must come from wind and solar

02

 **Fastest-ever growth**

Wind + solar: history's fastest technologies

Combined wind + solar output (~4,400 TWh) now nearly matches hydropower. Solar costs fell 90%+ in a decade.

Momentum is real and compounding
Policy and investment can accelerate further

03

Stark regional disparities persist

From 98% (Norway) to <2% (Saudi Arabia)

Scandinavia and Latin America lead at 65–98%, while Middle East and Sub-Saharan Africa lag far behind. The transition is fundamentally unequal — and closing this gap is both a climate and an energy justice imperative.

04

Transport & heating still lag

Primary energy share (14%) vs electricity (30%)

Transport and heating are harder to decarbonize — still reliant on oil and gas. This gap must close through electrification (EVs, heat pumps) plus green hydrogen for hard-to-abate sectors.

05



The Path Forward:

"Accelerating the transition requires scaling wind and solar, closing regional access gaps, and decarbonizing transport and heating."

Source: Our World in Data | Ember (2026) | Energy Institute – Statistical Review of World Energy (2025)

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